DATABASE MANAGEMENT SYSTEM

Data Integrity Constraints

Primary keys, foreign keys, UNIQUE, CHECK and DEFAULT rules, with verification

Submitted By	
Name	Palle Sai Rithesh
Roll Number	2511021240013
Course	BCA Game Development
Subject	Database Management System
Assignment	Assignment 6 — Data Integrity Constraints
Project	Online Shopping System

Assignment 6 — requirements

- Implement appropriate primary keys and foreign keys for all relevant tables.
- Apply UNIQUE, CHECK, and DEFAULT constraints wherever required.
- Explain the purpose of the constraints used.
- Execute suitable SQL statements to verify that data integrity and relationships between tables are maintained.

01 TASK 1

Primary Keys and Foreign Keys

Keys were created with the tables in Assignment 5. This section recaps them and explains each delete rule.

Primary keys. Every table has a single-column primary key, which guarantees that each row can be identified and can never be NULL or duplicated. Nine tables use an `INT AUTO_INCREMENT` surrogate key. The pincodes table uses the six-digit pincode itself, because it is already unique and never changes.

Table	Primary key	Constraint name
customers	customer_id	pk_customers
pincodes	pincode	pk_pincodes
addresses	address_id	pk_addresses
categories	category_id	pk_categories
products	product_id	pk_products
orders	order_id	pk_orders
order_items	order_item_id	pk_order_items
payments	payment_id	pk_payments
shipments	shipment_id	pk_shipments
reviews	review_id	pk_reviews

Foreign keys. A foreign key makes the database refuse a child row whose parent does not exist, and decides what happens when a parent is deleted. Where no action is written, MySQL's default applies: the delete is **rejected** (RESTRICT behaviour).

Foreign key	Child column → parent	On delete	Reason
fk_addresses_customer	addresses.customer_id → customers	CASCADE	Saved addresses mean nothing without their customer
fk_addresses_pincode	addresses.pincode → pincodes	RESTRICT	A pincode in use must not be removed
fk_products_category	products.category_id → categories	RESTRICT	A category that still holds products cannot disappear
fk_orders_customer	orders.customer_id → customers	RESTRICT	Order history must survive; customers with orders cannot be deleted
fk_orders_address	orders.address_id → addresses	RESTRICT	The delivery address of an order must stay on record
fk_order_items_order	order_items.order_id → orders	CASCADE	Lines belong to one order and go with it
fk_order_items_product	order_items.product_id → products	RESTRICT	Sold products stay in the catalogue records
fk_payments_order	payments.order_id → orders	RESTRICT	Financial records must never vanish silently
fk_shipments_order	shipments.order_id → orders	RESTRICT	Delivery records must be kept
fk_reviews_customer	reviews.customer_id → customers	CASCADE	Reviews are personal data of the customer
fk_reviews_product	reviews.product_id → products	CASCADE	A review of a removed product has no use

02 TASK 2

UNIQUE, CHECK and DEFAULT Constraints

Added with ALTER TABLE so that structure (Assignment 5) and business rules (this assignment) stay separate.

2.1 UNIQUE constraints

```
ALTER TABLE customers   ADD CONSTRAINT uq_customers_email        UNIQUE (email);
ALTER TABLE categories  ADD CONSTRAINT uq_categories_name         UNIQUE (category_name);
ALTER TABLE shipments   ADD CONSTRAINT uq_shipments_tracking      UNIQUE (tracking_no);
ALTER TABLE order_items ADD CONSTRAINT uq_order_items_order_prod  UNIQUE (order_id, product_id);
ALTER TABLE reviews     ADD CONSTRAINT uq_reviews_customer_prod   UNIQUE (customer_id, product_id);
```

Constraint	Rule it enforces
uq_customers_email	Two customers cannot register with the same e-mail address (it is the login name)
uq_categories_name	Category names are not repeated
uq_shipments_tracking	A courier tracking number identifies exactly one shipment
uq_order_items_order_prod	A product appears only once in an order; to buy more, increase quantity
uq_reviews_customer_prod	A customer can review a product only once

2.2 CHECK constraints

```
ALTER TABLE products ADD CONSTRAINT chk_products_price    CHECK (price > 0);
ALTER TABLE products ADD CONSTRAINT chk_products_stock    CHECK (stock_qty >= 0);
ALTER TABLE orders ADD CONSTRAINT chk_orders_total        CHECK (total_amount >= 0);
ALTER TABLE orders ADD CONSTRAINT chk_orders_status
    CHECK (status IN ('Placed', 'Confirmed', 'Completed', 'Cancelled'));
ALTER TABLE order_items ADD CONSTRAINT chk_items_quantity CHECK (quantity > 0);
ALTER TABLE order_items ADD CONSTRAINT chk_items_price    CHECK (unit_price > 0);
ALTER TABLE payments ADD CONSTRAINT chk_payments_amount   CHECK (amount > 0);
ALTER TABLE payments ADD CONSTRAINT chk_payments_method
    CHECK (payment_method IN ('UPI', 'Card', 'Net Banking', 'COD'));
ALTER TABLE payments ADD CONSTRAINT chk_payments_status
    CHECK (payment_status IN ('Pending', 'Paid', 'Failed', 'Refunded'));
ALTER TABLE shipments ADD CONSTRAINT chk_shipments_status
    CHECK (delivery_status IN ('Packed', 'Shipped', 'Out for delivery', 'Delivered'));
ALTER TABLE reviews ADD CONSTRAINT chk_reviews_rating     CHECK (rating BETWEEN 1 AND 5);
```

Constraint	Rule it enforces
chk_products_price, chk_items_price, chk_payments_amount	Money values must be greater than zero
chk_products_stock, chk_orders_total	Stock and order totals can be zero but never negative
chk_items_quantity	At least one unit must be ordered
chk_orders_status	Order status is one of Placed, Confirmed, Completed, Cancelled
chk_payments_method, chk_payments_status	Only the supported payment methods and payment states are accepted
chk_shipments_status	Delivery status follows the courier stages: Packed, Shipped, Out for delivery, Delivered
chk_reviews_rating	A rating is a whole number from 1 to 5

2.3 DEFAULT constraints

```
ALTER TABLE products    MODIFY stock_qty       INT         NOT NULL DEFAULT 0;
ALTER TABLE addresses   MODIFY label           VARCHAR(20) NOT NULL DEFAULT 'Home';
ALTER TABLE orders      MODIFY order_date      DATETIME    NOT NULL DEFAULT CURRENT_TIMESTAMP;
ALTER TABLE orders      MODIFY status          VARCHAR(20) NOT NULL DEFAULT 'Placed';
ALTER TABLE order_items MODIFY quantity        INT         NOT NULL DEFAULT 1;
ALTER TABLE payments    MODIFY payment_status  VARCHAR(30) NOT NULL DEFAULT 'Pending';
ALTER TABLE shipments   MODIFY delivery_status VARCHAR(30) NOT NULL DEFAULT 'Packed';
```

Column	Default	Why
products.stock_qty	0	A newly listed product has no stock until it is received

Column	Default	Why
addresses.label	'Home'	Most addresses are home addresses
orders.order_date	CURRENT_TIMESTAMP	The order time is recorded automatically
orders.status	'Placed'	Every order starts as Placed
order_items.quantity	1	One unit unless stated otherwise
payments.payment_status	'Pending'	A payment is pending until the gateway confirms it
shipments.delivery_status	'Packed'	A shipment record is created when the parcel is packed

> **Note:** CHECK constraints are enforced from **MySQL 8.0.16**; older versions accept the syntax but ignore the rule. Adding a constraint to a table that already holds rows validates those rows first, so the ALTER statements fail if existing data breaks a rule.

2.4 Confirming the constraints exist

```
SELECT CONSTRAINT_TYPE, COUNT(*) AS constraint_count
FROM information_schema.TABLE_CONSTRAINTS
WHERE TABLE_SCHEMA = 'online_shop'
GROUP BY CONSTRAINT_TYPE
ORDER BY CONSTRAINT_TYPE;
```

OUTPUT

CONSTRAINT_TYPE	constraint_count
CHECK	11
FOREIGN KEY	11
PRIMARY KEY	10
UNIQUE	5

(4 rows)

```
DESCRIBE orders;
```

OUTPUT

Field	Type	Null	Key	Default	Extra
order_id	int	NO	PRI	NULL	auto_increment
customer_id	int	NO	MUL	NULL	
address_id	int	NO	MUL	NULL	
order_date	datetime	NO		CURRENT_TIMESTAMP	DEFAULT_GENERATED
status	varchar(20)	NO		Placed	
total_amount	decimal(10,2)	NO		NULL	

(6 rows)

03 TASK 3

Purpose of Each Constraint Type

Constraint	Problem it prevents	Example in this database
PRIMARY KEY	Duplicate or missing row identity	Two customers with the same customer_id
FOREIGN KEY	Orphan rows and broken relationships	An order for customer 99 who does not exist
NOT NULL	Missing mandatory data	A customer without a name
UNIQUE	Repeated values where only one is allowed	The same e-mail registered twice; the same product twice in one order
CHECK	Values outside the allowed range or list	A negative price; a rating of 7; the status ‘Shipped’ on an order
DEFAULT	Forgotten values and inconsistent starting states	An order without a status or date

04 TASK 4

Verifying Data Integrity

Rule-breaking statements must be rejected, and valid statements must succeed.

The tests run on the database after the Assignment 7 data is loaded, because they need existing parent rows. Error messages are shown as MySQL 8.0 reports them. After every rejected statement the data is unchanged, because a failed statement is rolled back as a whole.

A. Statements that must be rejected

4.1 UNIQUE: duplicate e-mail

```
INSERT INTO customers (name, email, phone)
VALUES ('Duplicate Test', 'aarav.sharma@example.com', '9000000000');
```

```
ERROR 1062 (23000): Duplicate entry 'aarav.sharma@example.com' for key 'customers.uq_customers_email'
```

4.2 UNIQUE: same product twice in one order

```
INSERT INTO order_items (order_id, product_id, quantity, unit_price)
VALUES (1, 1, 1, 1499.00);
```

```
ERROR 1062 (23000): Duplicate entry '1-1' for key 'order_items.uq_order_items_order_prod'
```

4.3 UNIQUE: reused tracking number

```
INSERT INTO shipments (order_id, tracking_no, carrier)
VALUES (13, 'TRK20260001', 'Delhivery');
```

```
ERROR 1062 (23000): Duplicate entry 'TRK20260001' for key 'shipments.uq_shipments_tracking'
```

4.4 CHECK: negative price

```
UPDATE products SET price = -50 WHERE product_id = 1;
```

```
ERROR 3819 (HY000): Check constraint 'chk_products_price' is violated.
```

4.5 CHECK: rating outside 1 to 5

```
INSERT INTO reviews (customer_id, product_id, rating, review_text)
VALUES (4, 1, 7, 'Out of range');
```

```
ERROR 3819 (HY000): Check constraint 'chk_reviews_rating' is violated.
```

4.6 CHECK: status not allowed on an order

```
UPDATE orders SET status = 'Shipped' WHERE order_id = 13;
```

```
ERROR 3819 (HY000): Check constraint 'chk_orders_status' is violated.
```

4.7 CHECK: unsupported payment method

```
INSERT INTO payments (order_id, payment_method, amount)
VALUES (13, 'Cheque', 3197.00);
```

```
ERROR 3819 (HY000): Check constraint 'chk_payments_method' is violated.
```

4.8 CHECK: zero quantity

```
INSERT INTO order_items (order_id, product_id, quantity, unit_price)
VALUES (13, 8, 0, 599.00);
```

```
ERROR 3819 (HY000): Check constraint 'chk_items_quantity' is violated.
```

4.9 NOT NULL: missing mandatory column

```
INSERT INTO customers (email) VALUES ('no.name@example.com');
```

```
ERROR 1364 (HY000): Field 'name' doesn't have a default value
```

4.10 FOREIGN KEY: order for a customer that does not exist

```
INSERT INTO orders (customer_id, address_id, total_amount)
VALUES (99, 1, 500.00);
```

```
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails
(`online_shop`.`orders`, CONSTRAINT `fk_orders_customer` FOREIGN KEY (`customer_id`) REFERENCES `customers`
(`customer_id`))
```

4.11 FOREIGN KEY: deleting a category that still has products

```
DELETE FROM categories WHERE category_id = 1;
```

```
ERROR 1451 (23000): Cannot delete or update a parent row: a foreign key constraint fails
(`online_shop`.`products`, CONSTRAINT `fk_products_category` FOREIGN KEY (`category_id`) REFERENCES
`categories` (`category_id`))
```

4.12 FOREIGN KEY: deleting an order that has a payment

```
DELETE FROM orders WHERE order_id = 8;
```

```
ERROR 1451 (23000): Cannot delete or update a parent row: a foreign key constraint fails
(`online_shop`.`payments`, CONSTRAINT `fk_payments_order` FOREIGN KEY (`order_id`) REFERENCES `orders`
(`order_id`))
```

Note: All twelve statements were rejected and the tables still hold 12 customers, 15 orders, 30 order lines and 16 payments.

B. Statements that must succeed (defaults and cascades)

4.13 DEFAULT: status and date filled in automatically

```
START TRANSACTION;
INSERT INTO orders (customer_id, address_id, total_amount) VALUES (12, 13, 599.00);
SELECT order_id, status, order_date, total_amount FROM orders WHERE order_id = LAST_INSERT_ID();
ROLLBACK;
```

OUTPUT

order_id	status	order_date	total_amount
16	Placed	(time of insert)	599.00

(1 row)

Explanation: Neither status nor order_date was supplied. The defaults wrote *Placed* and the current date and time. ROLLBACK removes the test row.

4.14 DEFAULT: payment starts as Pending

```
START TRANSACTION;
INSERT INTO payments (order_id, payment_method, amount) VALUES (13, 'UPI', 3197.00);
SELECT payment_id, order_id, payment_method, payment_status FROM payments WHERE payment_id = LAST_INSERT_ID();
ROLLBACK;
```

OUTPUT

payment_id	order_id	payment_method	payment_status
17	13	UPI	Pending

(1 row)

4.15 ON DELETE CASCADE: deleting an order removes its lines

```
START TRANSACTION;
INSERT INTO orders (customer_id, address_id, total_amount) VALUES (12, 13, 599.00);
SET @oid = LAST_INSERT_ID();
INSERT INTO order_items (order_id, product_id, quantity, unit_price) VALUES (@oid, 8, 1, 599.00);
DELETE FROM orders WHERE order_id = @oid;
SELECT COUNT(*) AS lines_left FROM order_items WHERE order_id = @oid;
ROLLBACK;
```

OUTPUT

lines_left
0

(1 row)

Explanation: The test order had one line. Deleting the order also deleted the line, so no orphan row remains.

4.16 ON DELETE CASCADE: deleting a customer removes saved addresses

```
START TRANSACTION;
DELETE FROM customers WHERE customer_id = 12;   -- customer 12 has no orders
SELECT COUNT(*) AS addresses_left FROM addresses WHERE customer_id = 12;
ROLLBACK;
```

OUTPUT

addresses_left
0

(1 row)

Explanation: Customer 12 has no orders, so the delete is allowed and her saved address is removed with her. A customer who has orders cannot be deleted because of the restrictive foreign key on orders.

C. Consistency checks on the stored data

The eight orphan and total checks from Assignment 7 (section 4) all return 0. They prove that every foreign key has a parent and that orders.total_amount equals the sum of its order lines, a rule the CHECK constraints cannot express on their own.

Note: Conclusion. Primary keys, foreign keys, UNIQUE, CHECK, DEFAULT and NOT NULL rules together stop invalid data at the door: duplicate identities, orphan rows, impossible values and incomplete rows were all rejected, while valid operations and cascades behaved as designed.